UNIVERSITY OF CAPE TOWN

Test

Investment Decision Theory and Application
ECO4102Z/ECO4122Z

Time: 2 hours 30 minutes

Maximum Marks: 85

INSTRUCTIONS TO CANDIDATES

1. You have 5 minutes at the start of the examination in which to read the questions. You are strongly encouraged to use this time for reading only, but notes may be made. on this examination paper. You then have two and half ($2\frac{1}{2}$)hours to complete the paper.
2. You must not start writing your answers in the booklet until instructed to do so by the invigilator.
3. Mark allocations are shown in brackets [ ].
4. Attempt all 6 questions, beginning your answer to each full question on a new page.
5. Show all calculations where this is appropriate.
6. Write legibly using blue or black ink.
7. Formulae Sheet attached at the end of this question paper

1. Bheki and Melisa Jacobs, South African residents, are reviewing their financial plan. The Jacobs, both 53 years old, have one daughter, 18 years old. With their combined after-tax salaries totaling R1,800,000 a year, they are able to meet their living expenses and save R450,000 after taxes annually. They expect little change in either their incomes or expenses on an inflation-adjusted basis other than the addition of their daughter's college expenses. Their only long-term financial goal is to provide for themselves and for their daughter's education. The Jacobs both wish to retire in 10 years.

   Their daughter, a talented musician, is now entering an exclusive five-year college program. This program requires a R900,000 contribution, payable now, to the college's endowment fund. Thereafter, her tuition and living expenses, to be paid entirely by the Jacobs , are estimated at R720,000 annually.

   The Jacobs's personal investments total R20,800,000, and they plan to continue to manage the portfolio themselves. They prefer "conservative growth investments with minimal volatility." One-third of their portfolio is in the stock of Melisa's employer, a publicly traded technology company with a highly uncertain future. The shares have a very low-cost basis for tax purposes. The Jacobs, currently taxed at 30 percent on income and 20 percent on net realized capital gains, have accumulated losses from past unsuccessful investments that can be used to fully offset R1,800,000 of future realized gains.

   In 10 years, Bheki will receive a distribution from a family trust. His portion is now R21.6 million and is expected to grow prior to distribution. Bheki receives no income from the trust and has no influence over, or responsibility for, its management. The Jacobs know that these funds will change their financial situation materially but have excluded the trust from their current financial planning.

   (a) Construct the objectives and constraints portion of an investment policy statement for the Jacobs, addressing each of the following; in detail:

      i. Return objective
      ii. Risk tolerance
      iv. Liquidity requirements
      vi. Unique circumstances

   [12 marks]

   Ten years have passed. The Jacobs, now both aged 63, will retire this year. The distribution from Bheki's family trust will occur within the next two weeks. The Jacobs' current circumstances are summarized below:

   Personal Circumstances and Assets

   - Pension income will total R1,800,000 a year and will not increase with inflation.
   - Annual expenses will total R3,240,000 initially and will increase with inflation.
   - Inflation is expected to be 2 percent annually.
   - Their personal investments now total R18,000,000 (excluding trust distribution).

Question 1 continues on the next page...

- The Jacobs will rely on this R18,000,000 portfolio to support their lifestyle and do not wish to reduce their level of spending.
- The Jacobs have health problems and neither is expected to live more than 10 years. All health care expenses will be covered by employer-paid insurance.
- The Jacobs' daughter is now financially independent, and the Jacobs' sole investment objective is to meet their spending needs.
- The Jacobs are not concerned with growing or maintaining principal. The income deficit may be met with both investment income and by invading principal.

Trust Distribution Assets

- The trust distribution totals R36,000,000 and will occur within the next two weeks. No tax liability is created by the distribution.
- The Jacobs will maintain separate accounts for their personal assets and the trust distribution.
- They do not plan to withdraw income or principal.
- Tax liabilities produced by these assets will be paid from this portfolio.
- The Jacobs plan to donate these assets to an arts society when the surviving spouse dies. They have made a minimum pledge of R46.8 million toward construction of a new building.
- An after-tax annual return of 5.4 percent is required over five years to meet the minimum pledge.
- The Jacobs are concerned only that a minimum gift of R46.8 million is available. The Jacobs assume that at least one of them will live at least five years and that neither will live more than 10 years.

Alternative portfolios for the Jacobs' consideration appear in Table 1 below:

Table 1: Alternative Allocations

	Portfolio			
Asset Allocation	A	B	C	D
Domestic large-cap stocks	14%	30%	40%	30%
Domestic small-cap stocks	3	5	10	25
Foreign stocks	3	5	10	25
Intermediate-term fixed income	70	60	30	20
Cash equivalents	10	0	10	0
Total	100%	100%	100%	100%
Expected annual return*	4.2%	5.8%	7.5%	8.5%
Annual standard deviation	6.0%	8.0%	13.0%	18.0%

*Nominal after-tax returns

(b) For this part, answer either (i.) or (ii.), **do not answer both.**

(i.) Select and justify with three reasons the most appropriate of the four portfolios from the table above as an asset allocation for the Jacobs' R18,000,000.

(ii.) Select and justify with three reasons the most appropriate of the four portfolios from the table above as an asset allocation for the Jacobs' R36,000,000 in trust distribution assets.

[5 marks]

[**Total: 17 marks**]

The Jacobs' investment policy statement should include the following objectives and constraints.

i. Return objective: The Jacobs' return objective should reflect a total return approach that combines **capital appreciation and capital preservation**. After retirement, they will need approximately R1,350,000 (adjusted for inflation) annually to maintain their current standard of living. Given the Jacobs' limited needs and asset base, preserving their financial position on an **inflation-adjusted basis** may be a sufficient objective. Their long life expectancy and undetermined retirement needs, however, lead to the likely requirement for some asset growth over time, at least to counter any effects of inflation.

Although the Jacobs wish to exclude the future trust distribution from their current planning, that distribution will substantially increase their capital base and dramatically alter the return objective of their future IPS, primarily by reducing their needed return level.

ii. RiskTolerance:The Jacobs are in the **middle stage** of the investor lifecycle. Their income (relative to expenses), total financial resources, and long time horizon give them the ability to assume at least **an average, if not an above-average, level of investment risk.** Their stated preference of "minimal volatility" investments, however, apparently **indicates a below-average willingness to assume risk.** The large realized losses they incurred in previous investments may contribute to their desire for safety. Also, their need for continuing **cash outflow to meet their daughter's college expenses** may temporarily and slightly reduce their risk-taking ability. In sum, the Jacobs' risk tolerance is average.

Two other issues affect the Jacobs' ability to take risk. First, the holding of Melisa's company stock represents a large percentage of the Jacobs' total investable assets and thus is an important risk factor for their portfolio. Reducing the size of this holding or otherwise reducing the risk associated with a single large holding should be a priority for the Jacobs. Second, the future trust distribution will substantially increase their capital base and thus increase their ability to assume risk.

iii. Time Horizon: Overall, the Jacobs' ages and long life expectancies indicate a long time horizon. They face a **multistage horizon**, however, because of their changing cash flow and resource circumstances. Their time horizon can be viewed as having three distinct stages: the next five years from now (some assets, negative cash flow because of their daughter's college expenses), the following five years (some assets, positive cash flow), and beyond 10 years (increased assets from a sizable trust distribution, decreased income because they plan to retire). iv. Liquidity: The Jacobs need R900,000 now to contribute to the college's endowment fund. Alternatively, they may be able to contribute R900,000 of Melisa's low-cost-basis stock to meet the endowment obligation. In addition, they expect the regular annual college expenses (R720,000) to exceed their normal annual savings (R450,000) by R270,000 for each of the next five years. This relatively low cash flow requirement of 1.357 percent (270,000/19,900,000 asset base after R900,000 contribution, i.e 20,800,000 minus 900,000=19,900,000) can be substantially met through income generation from their portfolio, further reducing the need for sizable cash reserves. Once their daughter completes college, the Jacobs' liquidity needs should be minimal until retirement because their income more than adequately covers their living expenses.

v. Tax concerns: The Jacobs are subject to a 30 percent marginal tax rate for ordinary income and a 20 percent rate for realized capital gains. The difference in the rates makes investment returns in the form of capital gains preferable to equivalent amounts of taxable dividends and interest.

Although taxes on capital gains would normally be a concern to investors with low-cost-basis stock, this is not a major concern for the Jacobs because they have a tax loss carryforward of R1,800,000. The Jacobs can offset up to R1,800,000 in realized gains with the available tax loss carry forward without experiencing any cash outflow or any reduction in asset base.

vi. Unique Circumstances: The large holding of the low-basis stock in Melisa's company, a "technology company with a highly uncertain future," is a key factor to be included in the evaluation of the risk level of the Jacobs' portfolio and the future management of their assets. In particular, the family should systematically reduce the size of the investment in this single stock. Because of the existence of the tax loss carry forward, the stock position can be reduced by at least 50 percent (perhaps more depending on the exact cost basis of the stock) without reducing the asset base to pay a tax obligation.

In addition, the trust distribution in 10 years presents special circumstances for the Jacobs, although they prefer to ignore these future assets in their current planning. The trust will provide significant assets to help meet their long-term return needs and objectives. Any long-term investment policy for the family must consider this circumstance, and any recommended investment strategy must be adjusted before the distribution takes place.

Personal Portfolio: Portfolio A is the most appropriate portfolio for the Jacobs. Because their pension income will not cover their annual expenditures, the shortfall will not likely be met by the return on their investments, so the 10 percent cash reserve is appropriate. As the portfolio is depleted over time, it may be prudent to allocate more than 10 percent to cash equivalents. The income deficit will be met each year by a combination of investment return and capital invasion. Now that their daughter is financially independent, the Jacobs' sole objective for their personal portfolio is to provide for their own living expenses. Their willingness and need to accept risk is fairly low. Clearly, there is no need to expose the Jacobs to the possibility of a large loss. Also, their health situation has considerably shortened their time horizon. Therefore, a 70 percent allocation to intermediate-term high-grade fixed-income securities is warranted. The income deficit will rise each year as the Jacobs' expenses rise with inflation, but their pension income remains constant. The conservative 20 percent allocation to equities should provide diversification benefits and some protection against unanticipated inflation over the expected maximum 10-year time horizon.

Portfolio B, the second-best portfolio, has no cash reserves, so it could not meet the Jacobs' liquidity needs. Also, although it has a higher expected return, Portfolio B's asset allocation results in a somewhat higher standard deviation of returns than Portfolio A. Portfolios C and D offer higher expected returns but at markedly higher levels of risk and with relatively lower levels of current income. The Jacobs' large income requirements and low risk tolerance preclude the use of Portfolios C and D.

Trust Distribution Portfolio: Portfolio B is the most appropriate for the trust assets. Portfolio B's expected return of 5.8 percent exceeds the required return of 5.4 percent, and the required return will actually decline if the surviving spouse lives longer than five years. The portfolio's time horizon is relatively short, ranging from a minimum of 5 years to a maximum of 10 years. The Jacobs' sole objective for this money is to adequately fund the building addition. The portfolio's growth requirements are modest, and the Jacobs have below-average willingness to accept risk. The portfolio would be unlikely to achieve its objective if large, even short-term losses were absorbed during the minimum five-year time horizon. Except for taxes, no principal or income disbursements are expected for at least five years; therefore, only a minimal or even zero cash reserve is required. Accordingly, an allocation of 40 percent to equities to provide some growth and 60 percent to intermediate-term fixed-income to provide stability and capital preservation is appropriate.

There is no second-best portfolio. Portfolio A's cash level is higher than necessary, and the portfolio's expected return is insufficient to achieve the R46,800,000 value within the minimum value in five years.

Portfolio C has asufficient expected return, but it has a higher cash level than is necessary and, more importantly, a standard deviation of return that is too high given the Jacobs' below-average risk tolerance. Portfolio D has a sufficient return and an appropriate cash level but a clearly excessive risk (standard deviation) level. Portfolios C and D share the flaw of having excessive equity allocations that fail to recognize the relatively short time horizon and that generate risk levels much higher than necessary or warranted.

2. (a) Explain clearly the difference between a float-adjusted market-capitalization-weighted index and a market-value-weighted index. [2 marks]

   See notes

   (b) State and explain any four(4) categories of shares that are included in the free float restrictions when calculating the FTSE/JSE Africa Index Series. [4 marks]

   See notes

   Consider the three stocks in the following table. P_t represents price, in Rand, at time t, and Q_t represents shares outstanding at time t.

	P_t			Q_t		
Day	Alpha	Beta	Kappa	Alpha	Beta	Kappa
09 October	12	23	52	500	350	250
10 October	10	22	55	500	350	250
11 October	14	40	52	500	175*	250
12 October	13	47	25	500	175	500*
13 October	12	45	26	500	175	500

*There was a stock split at the close of 11 October and at the close of 12 October

(c) Calculate the rate of return on a price-weighted index of the three stocks for 09 October through 13 October.

[12 marks]

Treated as bonus question to be discussed in class

[**Total: 18 marks**]

3. (a) State the expected utility theorem. [1 marks]

A risk averse investor makes decisions using a quadratic utility function:

$$U(w) = w + dw^2.$$

(b) Derive an upper bound for d for this investor. [3 marks]

(c) Explain why the investor can only use this utility function to make decisions over a limited range of wealth, w. Your answer should include a statement of this range. [4 marks]

The investor states that the upper limit of wealth where she can use this utility function is $w = R100,000$.

(d) Determine the value of d in the investor's utility function. [2 marks]

The investor wins a prize of R25,000 in a gameshow. She is then offered the opportunity to exchange this prize for a larger prize of R60,000 if she can answer one more question correctly. However, she will receive no prize at all if she gets the question wrong. She estimates her chances of answering the question correctly to be 50%.

(e) Determine whether the investor should take this opportunity to exchange. [5 marks]

[**Total: 15 marks**]

Suggested Solution

(a) The expected utility theorem states that a function, $U(w)$, can be constructed representing an investor's utility of wealth, w, at some future date. [1/2] Decisions are made on the basis of maximising the expected value of utility under the investor's particular beliefs about the probability of different outcomes. [1/2]- definitions may vary- but key words are **maximise expected utility of wealth**

Question 3 continues on the next page. . .

(b) $U'(w) = 1 + 2dw$, and [$\frac{1}{2}$] $U''(w) = 2d$. [$\frac{1}{2}$] Because the investor is risk averse, we must have $U''(w) < 0$(alternatively to satisfy the condition of diminishing marginal utility of wealth (risk aversion)). So we must have $d < 0$. [2]

(c) The condition of non-satiation requires $U'(w) > 0$. [$\frac{1}{2}$] Hence $1 + 2dw > 0$ and $w > –1/(2d)$ [$\frac{1}{2}$] So the quadratic utility function can only satisfy the condition of non-satiation over a limited range of w: Specifically $–1/(2d) < w < \infty$ [1]

(d) $100,000 = –1/2d$ [1] => $d = –1/200,000 = –0.000005$ [1]

(e) $U(25,000) = 25,000–0.000005 \times 25,000^2 = 21,875$ [1]
$E[U(\text{exchange})] = 0.5 \times U(60,000) + 0.5 \times U(0)$ [1]
$= 0.5 \times (60,000 - 0.000005 \times 60,000^2) = 21,000$ [1]
So the investor should not accept the opportunity to exchange... [1] ... because the expected utility of the exchange opportunity is lower than that of the prize. [1]

4. Consider the following investments:

Investment 1		Investment 2	
Payoff	Probability	Payoff	Probability.
1	0.25	3	0.33
7	0.50	5	0.33
12	0.25	8	0.34

(a) What type of preferences do investors need in order to be able to use first-order stochastic dominance as a selection tool? [1 mark]

(b) Using first-order stochastic dominance, which of these investment opportunities is preferred over the other by an investor with the appropriate preferences? Show your calculations. [4 marks]

(c) What type of preferences do investors need in order to be able to use second-order stochastic dominance as a selection tool? [1 mark]

(d) Based on second-order stochastic dominance, which of the two investment opportunities is preferred by the investor with the appropriate preferences? Show your calculations. [4 marks]

[Total: 10 marks]

Suggested Solution

- - FOSD or FSD- non-satiated- prefer more to less
- FSD: No dominance . Sign is not monotone
- SOSD or SSD- non satiated and risk averse
- There is no SSD as the sign is not monotone.

5. (a) State, with reasons, whether each of the following statement about portfolio diversification is correct.

i. Proper diversification can reduce or eliminate systematic risk.
ii. Diversification reduces the portfolio's expected return because it reduces a portfolio's total risk.
iii. As more securities are added to a portfolio, total risk typically would be expected to fall at a decreasing rate.
iv. The risk-reducing benefits of diversification do not occur meaningfully until at least 30 individual securities are included in the portfolio.

[4 marks]

(b) In the context of mean-variance portfolio theory, define the term *'efficient portfolio'*.
[1 mark]

(c) Consider three securities: Motus Motors with expected return of 14% and standard deviation of return of 6%, Aving Ltd with average return of 8% and standard deviation of return of 3%, and Zazol Oil with mean return of 20% and standard deviation of return of 15%. Furthermore, assume that the correlation coefficient between Motus Motors and Aving Ltd is 0.5, between Motus Motors and Zazol Oil is 0.2, and between Aving Ltd and Zazol Oil is 0.4. Finally, assume that the riskless lending and borrowing rate is 5%.

Determine the expected return and standard deviation of a portfolio of the three securities that maximizes the Sharpe ratio (i.e. optimal portfolio). Assume no restrictions to short selling. [10 marks]

(d) Investors can hold any portfolio consisting of the three securities above and the risk-free asset. In addition, suppose a mutual fund offers an expected return of 12% and a standard deviation of 6%. All investors have standard mean-variance preferences. Should any investor invest in this mutual fund? Explain clearly why or why not. [3 marks]

[Total: 18 marks]

$$\mathbf{V}=\begin{bmatrix}0,0036 & 0,0009 & 0,0018\\ 0,0009 & 0,0009 & 0,0018\\ 0,0018 & 0,0018 & 0,0225\end{bmatrix};\ \mathbf{V}^{-1}=\begin{bmatrix}370,370 & -370,370 & 0,000\\ -370,370 & 1693,122 & -105,820\\ 0,000 & -105,820 & 52,910\end{bmatrix};\ \mathbf{R}=\begin{bmatrix}0,14\\ 0,08\\ 0,2\end{bmatrix}$$

$$\mathbf{R}-\mathbf{R}_f\mathbf{e}=\begin{bmatrix}0,09\\ 0,03\\ 0,15\end{bmatrix};\ \mathbf{w}=\begin{bmatrix}0,7778\\ 0,0556\\ 0,1667\end{bmatrix};\ R_p=0,1467 \text{ or } 14.67\%,\ \sigma^2=0,00338,\ \sigma=0,05817$$
or 5,817%

The expected return from the mutual is lower than the efficient portfolio, however its standard deviation is higher; this implies that it does ot plot of the efficient frontier. Therefore, any non-satiated and risk averse investor should not invest in this mutual fund. Some students may use the Sharpe ratio and observe that the mutual fund has a lower Sharpe ratio.

6. Consider the following assets in a world where the capital asset pricing model holds. These are the only risky assets in the market.

 Question 6 continues on the next page. . .

Asset	Expected return (% p.a.)	Total value of assets in market (Rbn)	Beta
Risky asset A	3.5	20	1.5
Risky asset B	2.2	30	0.2
Risky asset C	4.4	10	2.4

(a) Calculate:

i. the risk-free rate of interest.

i. the expected return on the market portfolio.

[4 marks]

The standard deviation of the return on the market portfolio is 10%.

(b) Calculate the market price of risk. [1 mark]

The risk-free rate of interest now increases to 3% p.a.

(c) Explain why one or more of the figures in the table must change. [2 mark]

[Total: 7 marks]

There are different approaches to this. Note we have 3 equations and 2 unknowns. We can use any two of these or all 3 to solve for the unknown. Approaches my differ but the answer is the same.

(i) We know that $E(r_i) = r_f + \beta_i \times (Er_m - r_f)[\frac{1}{2}]$ So 3.5 = r_f + 1.5 * $(R_M - r_f)$ $[\frac{1}{2}]$

And 2.2 = r_f + 0.2 * $(R_M - r_f)$ $[\frac{1}{2}]$

Subtracting the second equation from the first:

3.5 - 2.2 = 1.3 * $(R_M - r_f)$ $[\frac{1}{2}]$

So $R_M - r_f = 1[\frac{1}{2}]$

Also subtracting 7.5 times the second equation from the first:

3.5 - 7.5 * 2.2 $= -6.5r_f$ $[\frac{1}{2}]$

So $r_f = 2\%$ $[\frac{1}{2}]$ And $R_M = 3\%$ $[\frac{1}{2}]$

One may use the fact that $R_m = \sum_{i=1}^{n} \omega_i R_i \Rightarrow 2/6(3.5) + 3/6(2.2) + 1/6(4.4) = R_m, \Leftrightarrow R_m = 3\%$

(ii) $MPR = (R_M - r_f)/\sigma_m$= (3%- 2%) / 10% = 0.1 [1]

(iii) Every asset must have an expected return at least as high as the risk-free rate $[\frac{1}{2}]$ So the return on asset B must increase to at least 3% $[\frac{1}{2}]$

Intuitively if the risk-free rate increases, then investors will require a higher return on risky assets as well $[\frac{1}{2}]$

If the market price of risk remains unchanged, then the returns on all assets might just increase by 1% $[\frac{1}{2}]$

Some students may decide to use the SML to illustrate this